

RESPONSE TO THE R36 REFEREE REPORT

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1. REVISION AND EVIDENCE BOUNDARY

The latest report identifies a decisive defect: the R36 review object did not contain the numerical objects asserted in its protocol. We have replaced that publication practice with a materialized scientific revision. R38 contains a new article, technical supplement, this response, a computation report, executable constructors, independent verifiers, exact proof objects, complete result tables, build records, and a lossless historical annex. Every headline result below comes from a fresh execution of the committed R38 source. We do not retrospectively certify the execution claims in the R35 or R36 protocols.

The scientific target has not been replaced by an easier primary cohort. All 42 maintenance configurations, their frozen installed rules, horizons 4, 8, and 12, and all-state/all-restart tolerances 0.01 and 0.05 are retained. The new deterministic outer bound and separately feasible deployments agree exactly in 24 initial-distribution objectives, including 12 with strictly positive intervention cost. Twenty-three of these equalities also hold for the complete cost-to-go functions. The other 18 deterministic intervals are reported, rather than rounded away. Randomized and history-conditioned objectives are addressed by separate theorems, an exact constrained frontier reference, and a solved randomized Markov continuum example. The original stopped-control target is retained with a killed-operator connection and its full earlier scalar proof.

1.0.0.1. *Review provenance.* The addressed report is `reviews/2026-09-24-econometrica-r36/referee_report.md`, on review commit `c81410887bca5c16975beabb2ca8d6be7f235a19`. It reviews commit `c6cb75418f4d903750d001265f36ff4d568a9330`; the last materialized article in that ancestry was R34 at `d0cfe1b17cc66cde393f4e2d1700270975bb7e30`. The new branch descends from the review commit, thereby retaining the report and every earlier file. The R34 report and the original sources/results inform this revision; no other branch is modified.

1.0.0.2. *Reading order.* The article gives the economic problem and the main theorem chain. The supplement supplies representation rules, full configuration tables, proofs, and retained stopped-control details. The computation report defines the input/output contracts and audit boundary. The machine-readable ledgers report exact fractions and hashes that cannot usefully fit in a printed table. All paths below are relative to `revisions/2026-09-24-r38/`, unless identified as repository-root files.

2. RESPONSES TO THE BLOCKING FINDINGS

2.1. *R36-F1: materialize the claimed revision*

The correction is an executed package, not a new protocol. The root submission objects are `ECTA_R38.pdf`, `SUPP_R38.pdf`, `RESPONSE_R38.pdf`, `COMPUTATION_R38.pdf`, and `HISTORY_R38.pdf`, with their TeX entry files. The source tree contains the paper

modules and replication scripts. The results directory contains 42 primary exact proof objects, 70 constrained-frontier objects, 12 adaptive local-LP objects, 21 support-function bundles, and 42 gap-function bundles, in addition to the sensitivity and nonlinear-policy archives. A canonical comparison checks 187 exact scientific objects across the development and publication executions. This comparison is not mislabeled independent mathematical verification: the separate SymPy implementations verify 42 primary and 70 frontier objects. The publication manifest records the source commit, execution run, file hashes, and actual completed build checks.

2.2. R36-F2: revise the actual article and supplement

The manuscript has been rewritten around the now-executed theorem chain. Section 4 introduces the deterministic outer restriction with approximate as well as exact operating values. Section 5 treats the randomized policy classes directly. Section 6 replaces a sampled local multiplier by a certified local frontier. Section 7 presents the new primary optima, the gap diagnostics, the 66 economic sensitivities, and the nonlinear experiment. Section 8 supplies the killed-operator connection to the retained stopped economy. The technical supplement contains all primary, sensitivity, frontier, local-LP, and nonlinear rows. Earlier valid results are retained in the repository and in the lossless historical annex; they are not silently removed or relabeled as newly executed results.

2.3. R36-F3: exact operating value and a deployable outer bound

Exact V is now explicitly a reference input whose construction is charged. It is not needed for the general deterministic lower-bound theorem. Given checked $L_t \leq V_t \leq U_t$, define

$$\mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}.$$

Every operating-feasible deterministic action lies in this set, because $L_t - T_t^a U_{t+1} \leq V_t - T_t^a V_{t+1} \leq \varepsilon$. The implementation-cost recursion on this outer class is therefore valid without exact access to V . Both the safe-witness and exact-value lower recursions are executed on all 42 configurations; their differences are reported. The theorem also gives monotone refinement of nested operating witnesses to the exact necessary-action relaxation. That limit remains a relaxation unless its minimizing policy is feasible or its value matches a separately feasible policy. We do not convert convergence of an outer representation into an unsupported global convergence assertion.

2.4. R36-F4: deterministic, randomized, and history-conditioned classes

Section 2 defines deterministic Markov policies, randomized Markov kernels, and history-conditioned policies with operating feasibility at each admissible decision history. Conditioning occurs before the current action lottery. Section 4 proves the deterministic lower bound for both deterministic Markov policies and deterministic history-conditioned policies with that stronger restart requirement. It does not apply actionwise screening to a lottery.

The distinction is substantive. Section 5 solves a two-period version with the same maintenance primitives, defer incumbent, and uniform initial distribution. The deterministic optimum is $3/2$. The randomized Markov optimum is approximately 1.3720075153 at tolerance 0.01 and 0.8600375764 at tolerance 0.05. Exact rational enclosures of the analytic logarithmic integral accompany these numbers. Randomized sufficiency of deterministic actions is therefore false in this model family. The 24 primary equalities are always described as deterministic optima for the

stated initial objective. The separate frontier study gives exact history-conditioned, fixed-restart optima; it is not presented as the primary randomized Markov optimum.

2.5. R36-F5: *specify the objective and meaning of exactness*

The primary objective is expected discounted implementation cost under the uniform distribution on $[0, 1]$, at date zero, using the deployed policy's own transitions. Its constraint is operating feasibility at every state and date. The proof objects contain rational lower and upper integrated costs, rational cost-to-go functions, a policy hash, lower-function hash, frozen-input hash, and exact all-restart feasibility checks. The printed supplement gives all 42 decimal intervals; `results/primary_exact_values.json` gives the corresponding fractions and hashes.

Twenty-four cases have exact integrated equality, but only 23 have function equality at every date and state. The exception is horizon 8, neural31003, tolerance $1/20$. The article and tables distinguish these scopes. An isolated-state discrepancy cannot be erased by an integral, and neither can a later-restart discrepancy. Exactness is decided in rational arithmetic, never by a decimal tolerance.

2.6. R36-F6: *direct constrained comparators*

Two direct comparators are added. First, the necessary-action cost recursion supplies a lower bound for the same primary deterministic objective, while a separately checked policy supplies its upper bound. Equality certifies the constrained optimum in the 24 reported cases, including the 12 positive-cost cases. This is not a comparison with unconstrained operating DP or with a deliberately restricted sufficient class.

Second, the continuation-frontier recursion forms action-specific Minkowski sums of feasible successor value–cost sets, convexifies across current actions, and clips the operating constraint at every decision history. It exactly solves 70 horizon-four, fixed-restart history-conditioned problems. A separate primal/supporting-line implementation verifies all 70. Twenty-four are strictly below the deterministic lower cost at the same restart. The two-period analytic calculation additionally compares deterministic and randomized Markov policies under the same uniform initial objective. These are stronger reference calculations, but their policy classes and initial distributions remain explicit: 70 Dirac objectives are not substituted for a uniform-initial Markov objective, and the remaining primary randomized Markov optima are not claimed solved.

2.7. R36-F7: *local LP accuracy and distinct gap sources*

The local three-action, one-inequality problem is solved by feasible pure vertices and two-action mixtures. Its dual candidates are zero and nonnegative intersections of action lines. The cell constructor handles the resulting rational primal frontier by exact quadratic inequalities over whole cells, including endpoint limits; isolated points receive exact point-LP solutions. Directed dyadic coefficient encoding bounds representation error. The proof supplies a discounted accumulation bound for approximation to the continuous local restart relaxation.

The experiments separate five sources rather than assigning one residual to all of them. On all 42 configurations, the global support study compares constructed versus exact operating witnesses at fixed prices, then eight versus fifteen global prices at fixed exact value. The deterministic study compares safe versus exact necessary sets and the inherited versus enriched upper portfolio, and records infeasibility of the necessary-set selector. The local-LP diagnostic separately replaces the eight-price local approximation at three tolerances and two operating-witness choices. Its floor is fixed at zero. Thus it measures local multiplier/representation error;

it is not an additive decomposition of the nonzero inherited R34 support floor, nor a proof that a structural global duality gap vanishes. Global prices λ and local dual variables μ use separate notation and result fields.

2.8. R36-F8: explain the 18 remaining deterministic intervals

The full lower-set selector is operating-feasible in all 14 horizon-four configurations. It attains the outer minimum there, explaining why all 14 close: no further upper search is needed to attain that lower bound. At the longer horizons, 19 necessary-set selectors fail the exact all-restart operating test. One of these configurations has a different feasible upper policy with the same initial integral, leaving the 18 nonzero intervals. This directly exhibits the accumulation of operating losses ignored by the pointwise necessary test.

For each residual case the ledger reports the necessary selector's operating violation, safe and exact outer bounds, inherited and new feasible upper costs, witness improvements, support-price improvements, and the remaining deterministic gap. The maximum remaining gap is approximately 0.103474 at horizon 8 and 0.248472 at horizon 12. These are not numerical zeroes. The reference calculation certifies its interval at every finite execution; it declares a solved optimum only on equality. The local-LP convergence theorem is not used to assert a stopping guarantee for these structural residuals.

2.9. R36-F9: execute the nonlinear multistate study

The new economy has two continuous condition states, four actions, two shocks, bilinear action-dependent transitions, and no exact one-dimensional affine closure. Whole-box integer enclosures use scale 2^{34} and directed rounding. The grid contains 16 through 256 cells per coordinate, and horizons are 4, 8, 16, and 32. Costs account conservatively for the installed rule's threshold changes inside each box. Array hashes, witness envelopes, candidate policies, timing, and memory/representation metadata are serialized.

All 76 candidate records are published: 40 restart-generator outputs, 20 classical four-price scalarization portfolios, and 16 compiled-rule pilot outputs. The restart generators obtain 26 operating certificates; the classical portfolios obtain 15; the compiled-rule pilot obtains none. A failed sufficient box certificate is not evidence that the underlying policy is truly infeasible. The separate operating tolerance is $1/2$, not either primary tolerance. The baseline uses the same primitives, grids, and final box evaluator; two restart penalties are not treated as independent extra instances of the classical comparator.

The pilot is a fresh R38 midpoint-compiled box policy, not an asserted reconstruction of an unavailable R35 execution. We retain its failures and distinguish it from the continuous installed rule on threshold-crossing boxes. The study is exploratory and does not claim a prospective protocol that did not exist. The full source and output now replace the earlier unsupported description of an extension.

2.10. R36-F10: economic sensitivities and calibration

All 66 specified combinations are executed: eleven one-at-a-time specifications, three condition/calendar installed rules, and two explicitly labeled tolerances, with horizon eight. The specifications vary discounting, implementation weights, initial density, maintenance cost, and defer persistence. Dynamic and pointwise revision use the same sufficient action restrictions, but each policy's own transition occupancy. Twenty-two have strictly positive dynamic savings, 44 have exactly zero savings, and all 66 pass the operating check. The largest normalized cost

saving is approximately 0.087011. The full table is retained, not only cases with an effect. These scenarios establish sensitivity of the mechanism beyond the deliberately engineered incumbent; they are not a random sample, an empirical frequency estimate, or dollar calibration.

2.11. *R36-F11: genuinely separate verification*

`replication/verify_primary.py` uses SymPy Rational and a separate row/partition parser. It imports neither the Fraction/PW constructor nor its composition, selection, or optimizing-envelope routines. It independently checks primitive values, transition pullbacks, necessary-action threshold crossings, endpoint limits, isolated points, terminal conditions, policy equations, lower recursions, exact integrals, and the scope of each equality. All 42 primary objects pass. Twenty categories of deliberately corrupted inputs are rejected, including model primitives, action labels, malformed rational strings, partition changes, metadata, integrals, and frozen-input hashes.

`replication/verify_frontier.py` separately checks all 70 fixed-restart frontiers using rational feasible combinations and supporting-line inequalities. This verifier shares the independent parser and literal model specification, but not the frontier constructor. Residual common-mode risk remains in the common mathematical specification, the Python runtime, and the declared serialization contract. The local-LP and nonlinear checks use their own constructor arithmetic and are called exact/read-only checks, not independent implementations. The computation report states those boundaries explicitly.

2.12. *R36-F12: publication discipline and provenance*

The publication process executes the scientific source, checks every headline count and all 187 canonical exact objects, and commits the results before the paper build. The manuscript tables are generated from those committed ledgers. A separate publication stage compiles the actual TeX entry points and records completed build checks and hashes. Its manifest identifies the scientific source commit and workflow run; the final repository commit identifies the complete review object. A workflow definition by itself is not treated as a successful run. The preserved R35/R36 protocols are historical assertions, not evidence replacing any of these new objects.

2.13. *R36-F13: neural proposals and the paper's identity*

The revised title is *Certified Bellman Operators for Costly Policy Revision*. The constructive contribution is the economically constrained revision problem and its certificates, not an unestablished neural speed advantage. All 18 frozen neural configurations remain in the primary cohort and all previous neural results remain accessible in the historical annex and repository. Their training is an amortized inherited input, not newly executed computation. Neural-specific performance is neither manufactured nor used to support the positive-cost optimality result. The paper retains neural proposals as one installed-rule family on the same terms as the classical alternatives.

2.14. *R36-F14: a method-level connection to stopped control*

The original stopped consumption–portfolio model and its all-domain operating target remain in the article. Section 8 constructs the relevant link through a positive killed transition operator, with exit reward separated from survival continuation. Supersolution and policy-comparison inequalities extend to its subprobability kernel; a supplementary error recursion separates

TABLE I
CHARGED PRIMARY CONSTRUCTION WORK AND REPRESENTATION

Horizon	Operating DP (s)	Median revision (s)	Max. lower pieces	Max. bits
4	0.0046	0.0973	11	143
8	0.0141	0.3478	90	186
12	0.1198	2.3529	430	242

Notes. The operating DP is executed once per horizon. Revision time includes both exact-value and safe-witness outer bounds and the complete upper portfolio. Independent verification, support diagnostics, and supplementary studies are additional work, not omitted overhead. Timings are generated from this publication execution.

quadrature/transition, representation, policy, and terminal errors. A continuous-action application additionally needs verified action coverage: a finite action grid without an approximation bound cannot supply an upper witness for the full control problem.

This is an operator-level connection, not a claim that maintenance results solve the stopped economy. The earlier scalar constant-adjustment optimum and its full derivative/survival proof are retained in the supplement. The previously reported all-domain stopped-control bound, 7.181834580823298, does not meet the original 0.01 target and is not relabeled as doing so. The lossless historical annex preserves the complete earlier research program, rather than deleting unsuccessful or auxiliary material.

3. RESPONSES TO TECHNICAL COMMENTS

3.0.0.1. *R36-T1: necessary-action lemma.* Section 4 states it as a theorem, specifies the policy classes and information timing, and proves both approximate-witness necessity and the cost lower recursion. It also states the exact integrated equality test and the stronger functionwise equality test separately.

3.0.0.2. *R36-T2: deterministic history dependence.* Section 2 requires feasibility after each admissible decision history. The lower recursion remains a bound because each chosen deterministic action belongs to the outer set and each history-specific continuation cost is bounded below by the statewise Bellman lower function. No Markovianity of the candidate history-dependent policy is assumed in that induction.

3.0.0.3. *R36-T3: exact reference values.* All 42 rational pairs are serialized in the primary ledger and the compact exact-value index. The table gives decimal renderings only for readability. Policy, raw-rule, lower-function, and complete proof hashes are included per row.

3.0.0.4. *R36-T4: horizon-four closure.* The exact necessary-set cost selector is itself feasible in all 14 cases. The exact operating-regret functions verify this statewise. The corresponding property fails for 19 longer-horizon configurations, which provides a concrete structural diagnostic rather than inferring causation solely from the number of solved cases.

3.0.0.5. *R36-T5: charge the exact operating solve.* Table I below reproduces the generated construction account. Exact operating DP is executed once per horizon; both outer recursions and the complete upper-candidate portfolio are charged per configuration. The independent audit, support diagnostics, and supplementary experiments are additional recorded stages. Bit lengths, piece counts, and serialized sizes are reported rather than treating exact arithmetic as unit cost.

3.0.0.6. *R36-T6: reference versus deployment.* The exact-value specialization is labeled reference certification. The safe L/U theorem and its executed bounds establish how the construction is used without an exact operating solve. No timing result for the special one-dimensional reference is called a high-dimensional deployment rate.

3.0.0.7. *R36-T7: local LP degeneracies.* The supplement lists feasible pure vertices, two-action mixtures, zero and nonnegative dual intersections, deterministic tie rules, denominator sign partitions, and exact isolated-point treatment. A zero denominator is not evaluated as a regular rational branch. If no candidate passes the full-cell checks, the constructor refines or reports a checked failure.

3.0.0.8. *R36-T8: adaptive whole-cell certificate.* The proof multiplies the primal–dual gap by a positive affine denominator and checks a quadratic at both endpoint limits and any interior vertex. The recorded dual function is verified against every action line, not merely at a midpoint. Coefficients are rounded down with an explicit 2^{-39} per-stage allowance. Conditional finite termination requires a stable feasible representation on one-sided cells; the imposed depth cap is a failure condition, not a proof of unconditional global convergence.

3.0.0.9. *R36-T9: distinct multipliers.* Global support prices λ and statewise local duals μ have different objectives, notation, source functions, and experimental fields. Optimizing the latter does not optimize the former or establish strong duality.

3.0.0.10. *R36-T10: gap diagnostics.* The 42-row gap ledger contains the two support-ladder steps, safe/exact deterministic witness step, upper-portfolio change, inherited restart floor, necessary-policy violation, and residual interval. The 12 local diagnostics are explicitly separate zero-floor experiments. We avoid a misleading additive decomposition across different policy classes or continuation floors.

3.0.0.11. *R36-T11: warm and cold costs.* Installed rules and historical fitting are frozen amortized inputs. Fresh operating solutions, witnesses, cost recursions, upper search, all verification, frontier construction, support refinement, nonlinear grids, and sensitivity calculations are charged in the corresponding ledger. The computation report separates stage-internal time from subprocess/dependency and publication overhead.

3.0.0.12. *R36-T12: primitives and parser risk.* Mutation tests alter probabilities, transitions, discounting, revision primitives, action identifiers, rational fields, partitions, endpoints, and scope metadata. The separate verifier compares literal primitives rather than trusting a constructor-supplied model label. Shared mathematical assumptions and remaining runtime/serialization dependencies are declared.

3.0.0.13. *R36-T13: unsuccessful nonlinear runs.* All 16 fresh compiled-rule pilot records and every failed restart/classical certificate remain in the same 76-row ledger and supplement table. The nonlinear tolerance and exploratory chronology are prominent. The earlier protocol's unavailable pilot is not manufactured as a new data record.

3.0.0.14. *R36-T14: designed sensitivities.* All eleven specifications and three installed-rule definitions are explicit; 22 effects and 44 zero effects are both reported. The paper makes no prevalence or calibration claim from a designed grid.

3.0.0.15. *R36-T15: qualified global terminology.* Every optimum is indexed by policy class and initial objective. Deterministic global equality, history-conditioned fixed-restart equality, randomized Markov two-period equality, and lower bounds over randomized policies are different objects throughout the manuscript and serialized metadata.

4. CONNECTION TO THE PRECEDING R34 REPORT

R34-F1/F2 and T2 asked for a same-object reference and positive-cost closure. The necessary-action equality certificates and constrained frontier reference supply new calculations rather than another presentation of the old sufficient class. R34-F3, F9, T3, T4, and T8 asked for principled multipliers and gap diagnosis. The exact local LP, whole-cell accuracy theorem, 15-price global study, and explicit upper-search ladder address those requests, with the remaining structural intervals retained.

R34-F4, T6, and T7 asked to leave the favorable one-dimensional affine setting. The nonlinear two-state experiment reaches horizon 32 and reports the full box arithmetic and a same-grid classical baseline. R34-F5/F8 asked for economically interpretable occupancy sensitivity. The 66-condition study varies primitives and natural installed rules and preserves null outcomes. R34-F6 asked to align neural emphasis with evidence; the revised title does so without discarding any neural cohort. R34-F10/T9 asked for a distinct arithmetic kernel; the SymPy primary and frontier verifiers supply that boundary.

R34-F7/F12 asked for an identifiable advance beyond assembling familiar inequalities. The new result is exact certification of strictly positive constrained implementation costs, together with a witness-based outer method and a verified local approximation procedure. We locate that contribution without claiming that classical Bellman comparison or weak duality is itself novel. R34-F11/T10 asked for coherence with the original stopped economy. The killed-operator comparison supplies a mathematical connection while the complete earlier evidence remains in the supplement and historical annex. R34-T1/T5 are addressed by the formal history classes and the consistent computational accounting.

5. WHAT THE COMPLETED EVIDENCE ESTABLISHES

The revision now contains inspectable positive deterministic optima, exact constrained randomized references for their declared objectives, a controlled local approximation theorem, broader economic and nonlinear tests, and independent proof checking. It is a materially different scientific object from the protocol-only R36 branch. The full primary randomized Markov optimum, the 18 residual primary deterministic gaps, empirical dollar calibration, neural-specific acceleration, and the original stopped economy's 0.01 all-domain target are not claimed to follow from these results. Keeping those distinctions explicit is part of the proof and computation, not a substitute for the constructive results above.